System Design for an Intelligent Hair Dryer

Submitted By: Zoya Bari (2720637)

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Submitted To: Dr. Naeem Shehzad

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Declaration of the Use of Generative AI

I declare that no outputs of generative AI such as ChatGPT, Gemini, Co-pilot or any other forums have been presented as my own work

Introduction

The patent US 12178303 B2, attributed to Spur Concepts Inc., describes an "Intelligent Hair Drying/Styling System" aimed at addressing the limitations of traditional dryers, which often cause thermal damage due to excessive heat and an inability to adapt to hair characteristics like curl pattern, density, and porosity. The device is built around a hybrid heating system that combines a Quartz Infrared Bulb with a Nichrome Wire Heater, all powered by a standard DC motor and fan. To achieve autonomous adaptation, the device employs a Contactless Infrared Sensor to monitor hair temperature and a 9-axis IMU to track movement, orientation and speed, ensuring the user does not dwell on one spot for too long and to adjust the flow, speed and concentration of the air via it's nozzle .The whole system is controlled by an ARM Cortex M3 processor running an RTOS, which handles Wi-Fi/Bluetooth syncing and manages a Lithium-Ion battery.

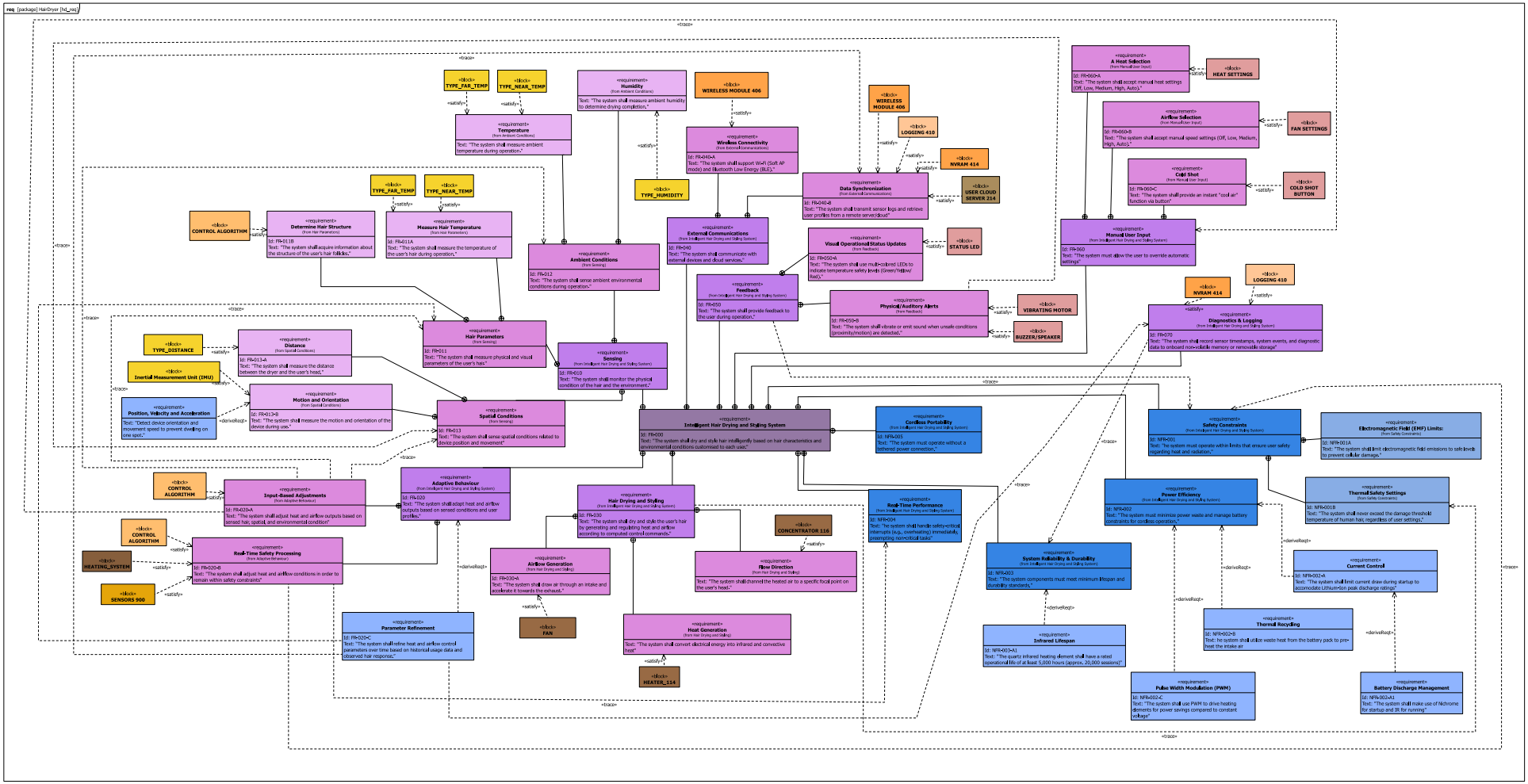
This report presents a SysML model of the device, capturing its requirements, structural architecture, and behaviour. The report consists of a Requirements Diagram, Block Definition Diagram (BDD), and Activity Diagram made using the online open-source software Gaphor version 3.2.0, utilising its components and features.

The Requirements Diagram defines system functions and links them to structural elements, illustrating containment, derived, verification, satisfactory and traceability relationships. I found it difficult to distinguish between some relationships, so my main modelling principle was that a containment relationship was used when the parent requirement cannot be satisfied unless all child requirements are satisfied. I also utilised colour coding to provide a clear visual representation of the hierarchy: Purple (Main Functional), Pink/Light Pink (Contained Sub-requirements), Blue (NFRs), Lighter Blue (Contained NFRs) and Baby Blue (Derived). Most requirements were extracted from the "Summary of the Invention" and "Detailed Description" and are satisfied by blocks in the BDD.

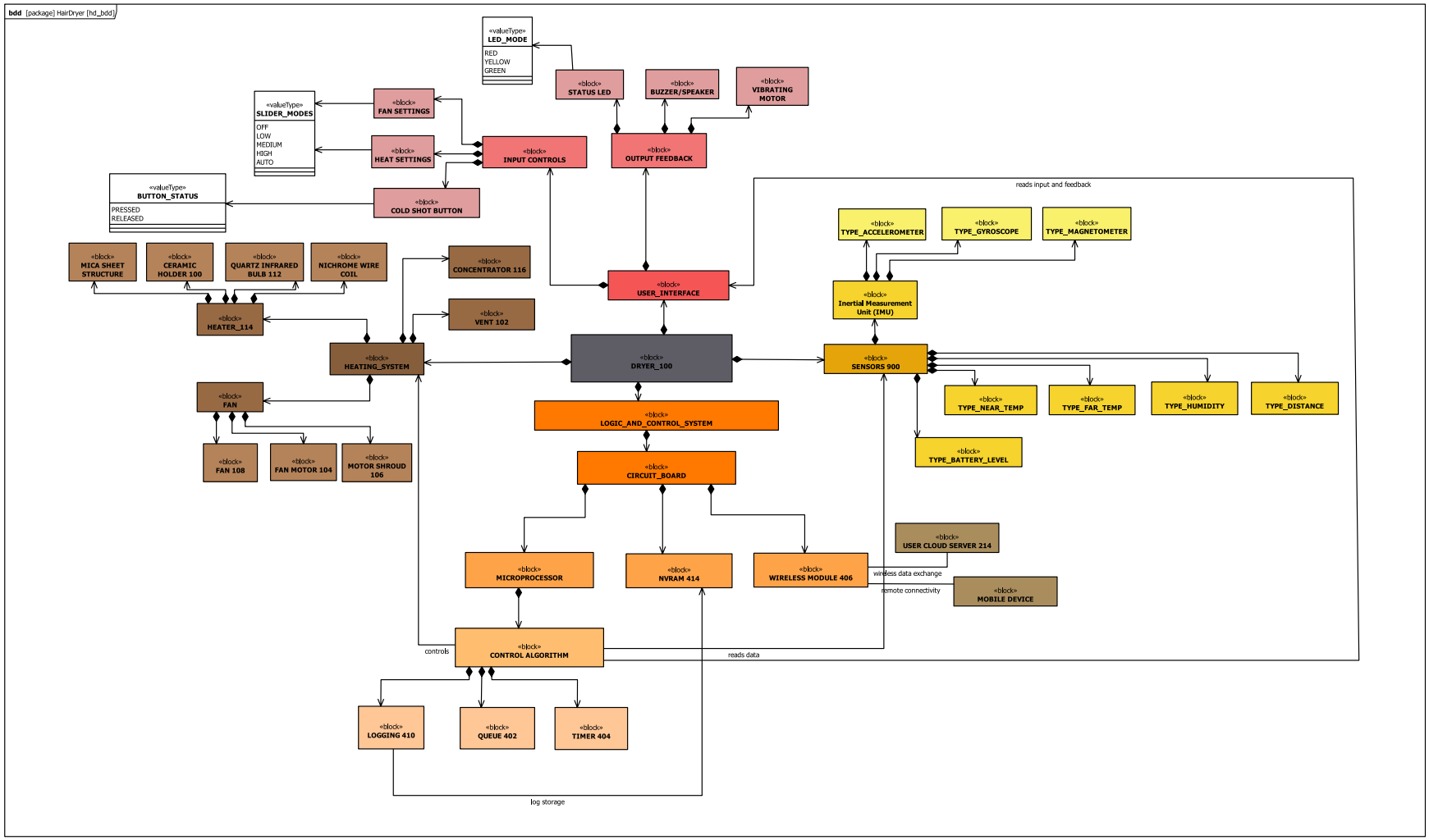
The Block Definition Diagram (BDD) acts as a structural blueprint, breaking down the system into component parts. Creating this was a rigorous process involving close analysis of the diagrams on Pages 2–16 as well as the written Brief Description & Detailed Description to identify every part. I color-coded the diagram, using different colours for separate subsystems and gradients of the same colour family to visually distinguish the top-level blocks from their constituent parts. In typical SysML v2 notation, certain dependency or usage relationships are represented by dotted lines with open arrows. However, the block diagram section of software I used (Gaphor) did not support this specific notation. To overcome this limitation, I made use of the direct association arrow to represent these connections. I inferred the relationships based on my understanding of how the system works. For example, I connected Logging to NVRAM to show where diagnostic data is stored and linked the Control Algorithm to the hardware (sensors, heating, UI) to represent the main loop where the processor reads data and controls the actuators.

The Activity Diagram models the behavioural logic of the Control Loop rather than a simple linear flow. I constructed this by synthesising the Summary of the Invention and analyzing FIG. 7 (Queue Structure) and FIG. 8 (Timer Module) on Pages 6–11. These figures were critical in understanding how the software manages tasks. The diagram illustrates how the Command Queue manages sensor and actuator adjustments in its operation. Swimlanes are used to allocate responsibility across Sensors, Control Algorithm, and Actuators, ensuring consistency with the structural decomposition presented in the BDD.

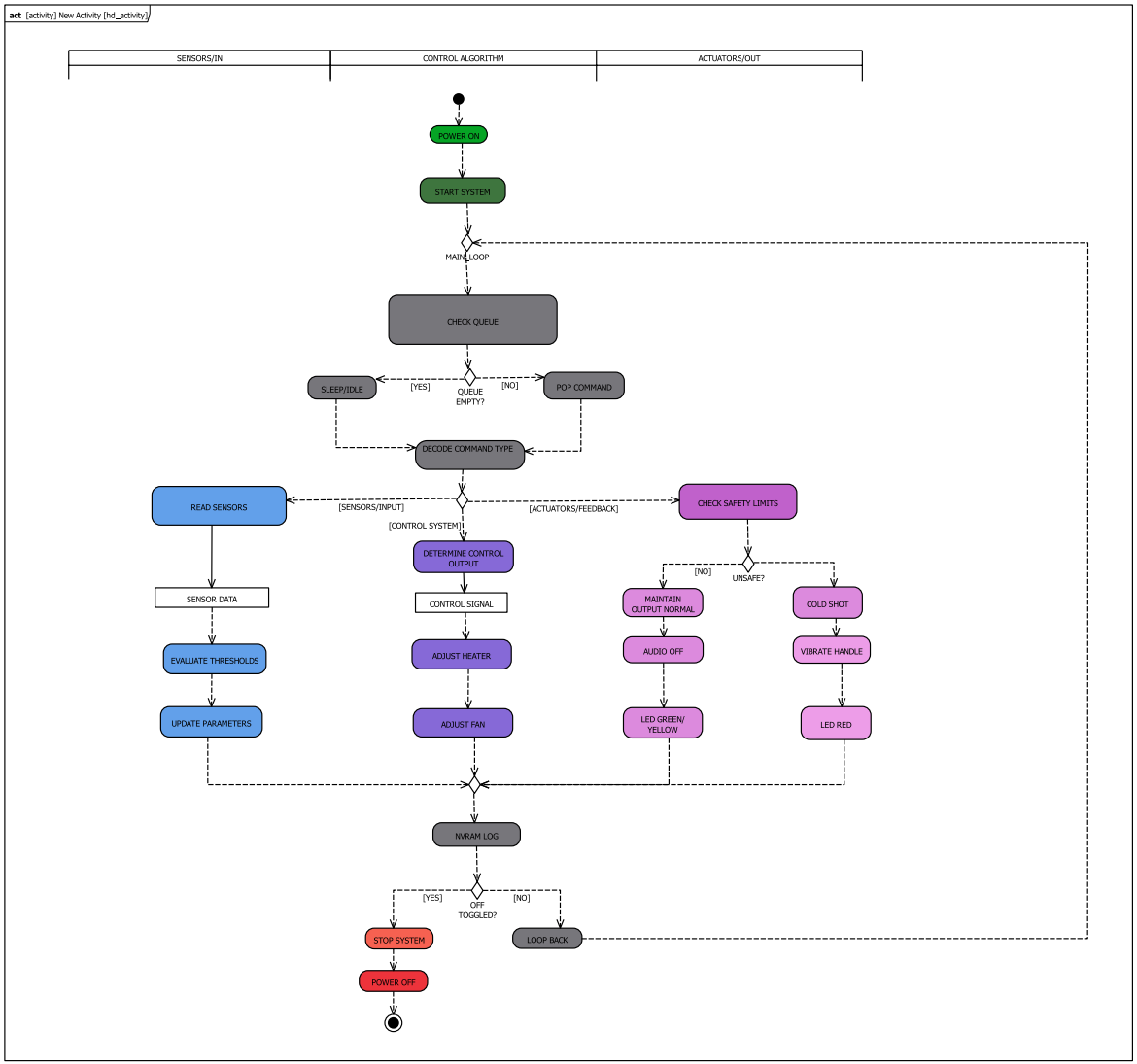
Requirements Diagram



Body Definition Diagram (BDD)



Activity Diagram



Appendix

References:

1. **Goldman, R., Friedman, J. and Rattner, Z.** (2024). *System and Method for an Enhanced Hair Dryer*. U.S. Patent 12,178,303 B2. Washington, DC: U.S. Patent and Trademark Office. Available at: https://patents.google.com/patent/US12178303B2

2. **Gaphor Developers**. *Gaphor Documentation (Version 3.2.0)*. [Software Documentation]. Available at: https://docs.gaphor.org